

Privacy-Preserving Disease Prediction Using Federated Learning

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Abstract

This paper presents a privacy-preserving disease prediction system using federated learning. Traditional machine learning approaches rely on centralized data, which can lead to privacy risks and data leakage. In this work, a federated learning approach is proposed where multiple clients train local models on decentralized healthcare data without sharing sensitive information. The global model is updated by aggregating local model parameters, ensuring data privacy while improving prediction accuracy. Experimental results using machine learning models such as Logistic Regression and Random Forest demonstrate that the proposed approach achieves reliable performance. The Random Forest model achieved the highest accuracy of approximately 95%, making it suitable for heart disease prediction. This work highlights the importance of privacy-aware machine learning in healthcare systems.

1. Introduction

In recent years, machine learning has been widely used in healthcare for disease prediction and diagnosis. However, traditional approaches require centralized data, which raises serious privacy concerns. Healthcare data is highly sensitive and cannot be easily shared across organizations. To address this issue, federated learning has emerged as a promising solution that enables collaborative model training without sharing raw data.

Heart disease is one of the leading causes of death worldwide, making early prediction and diagnosis crucial. Machine learning models can help in analyzing patient data and predicting the risk of heart disease. However, ensuring data privacy while maintaining

model accuracy remains a significant challenge. This paper proposes a privacy-preserving approach using federated learning combined with machine learning models to address this issue.

2. Literature Review

2.1 Federated Learning in Healthcare

Federated Learning (FL) has emerged as a promising approach for privacy-preserving machine learning in healthcare systems. Traditional machine learning techniques rely on centralized datasets, where sensitive patient data is collected and stored in a single location, increasing the risk of data breaches and privacy violations. To address this issue, Federated Learning enables decentralized training, where multiple clients collaboratively train a global model without sharing raw data.

Recent studies have shown that Federated Learning can achieve performance comparable to centralized models while maintaining data privacy. It allows healthcare institutions to collaborate without exposing sensitive patient information. However, challenges such as communication cost, data heterogeneity, and system scalability remain areas of active research.

2.2 Machine Learning for Heart Disease Prediction

Machine learning techniques have been widely applied in the field of healthcare for disease prediction and diagnosis. In particular, heart disease prediction has gained significant attention due to the increasing number of cardiovascular cases worldwide. Algorithms such as Logistic Regression, Decision Trees, Support Vector Machines, and Random Forest have been extensively used to analyze patient data and predict disease outcomes.

Among these models, Random Forest has shown superior performance due to its ability to handle large datasets and capture complex relationships between features. Studies indicate that proper data preprocessing, feature selection, and model optimization play a crucial role in improving prediction accuracy. However, the use of centralized data raises concerns regarding privacy and security.

3. Methodology

3.1 Dataset

The dataset used in this study consists of various health-related attributes such as BMI, age, income, physical health, mental health, and lifestyle factors. The target variable indicates whether a person has heart disease or not.

3.2 Data Preprocessing

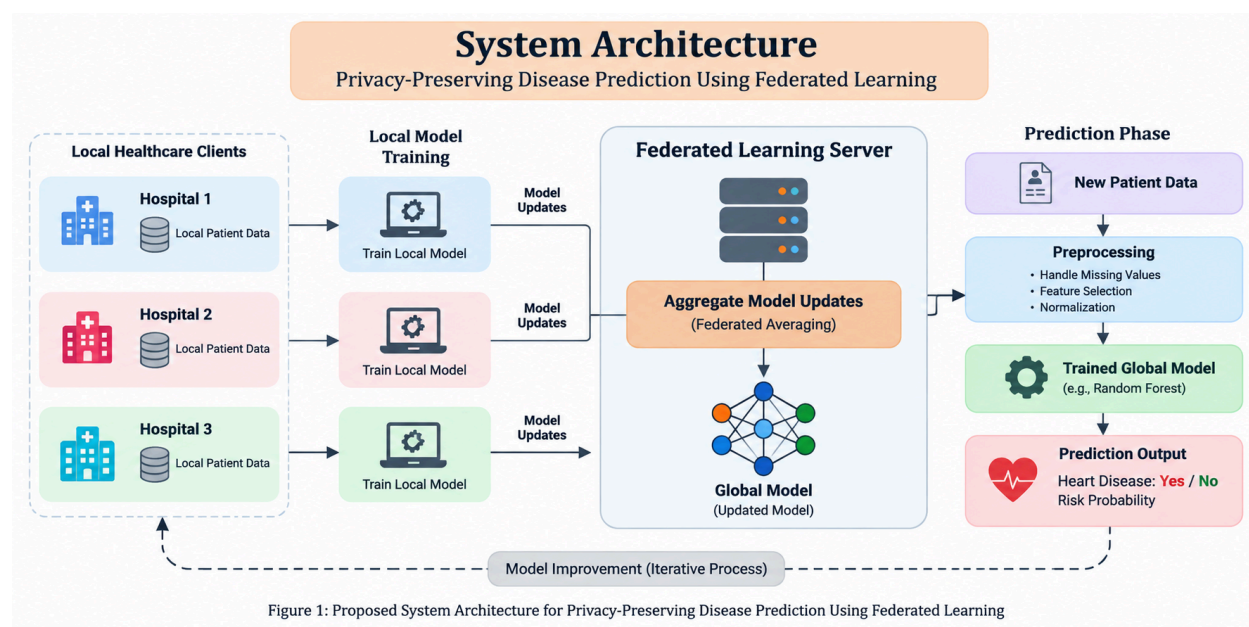
The dataset was cleaned and preprocessed by handling missing values and selecting relevant features. The data was then split into training and testing sets using an 80:20 ratio.

3.3 Model Selection

Two machine learning models were used in this study:

- Logistic Regression
- Random Forest Classifier

Figure 1: Proposed System Architecture for Privacy-Preserving Disease Prediction Using Federated Learning



As shown in Figure 1, the proposed system enables decentralized model training using federated learning while preserving data privacy.

3.4 Training and Evaluation

The models were trained on the training dataset and evaluated on the testing dataset. Performance metrics such as accuracy, precision, recall, and F1-score were used for evaluation.

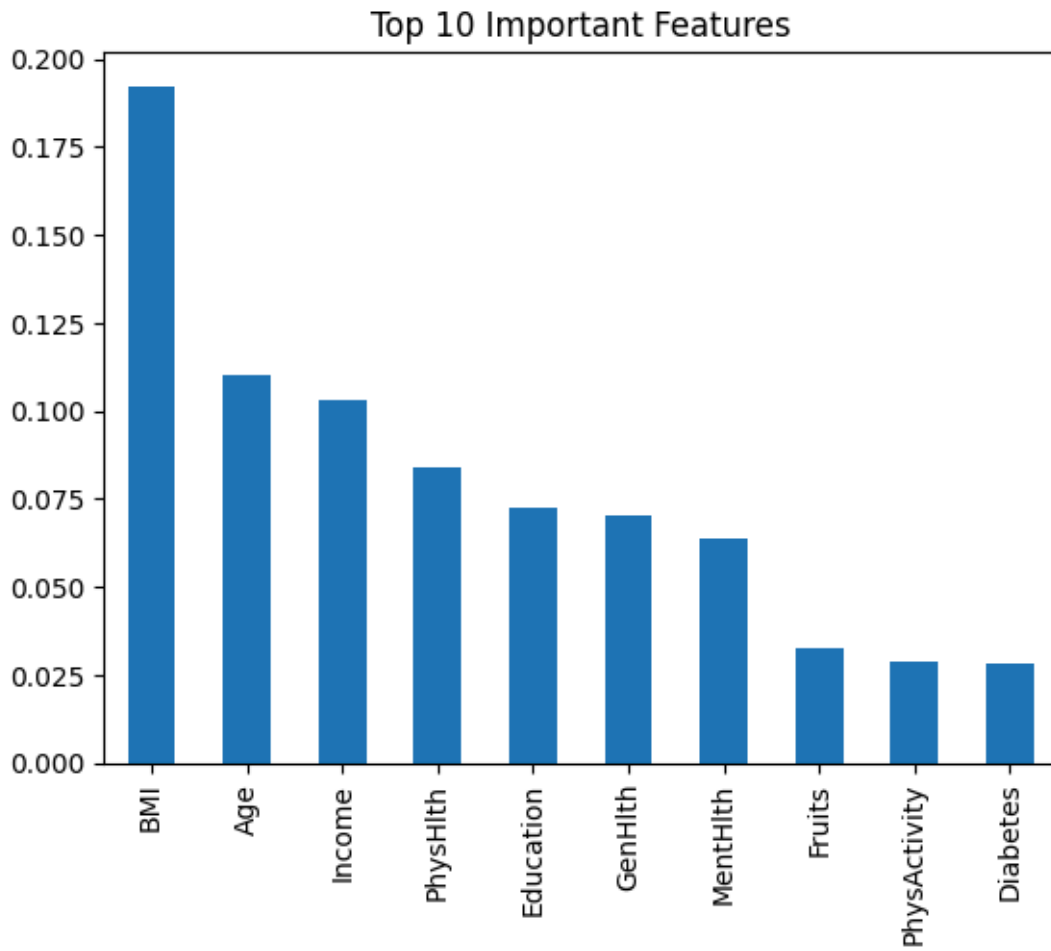
4. Results and Discussion

The performance of the models was evaluated using classification metrics.

- Logistic Regression achieved an accuracy of approximately **78%**.
- Random Forest achieved an accuracy of approximately **95%**, outperforming Logistic Regression.

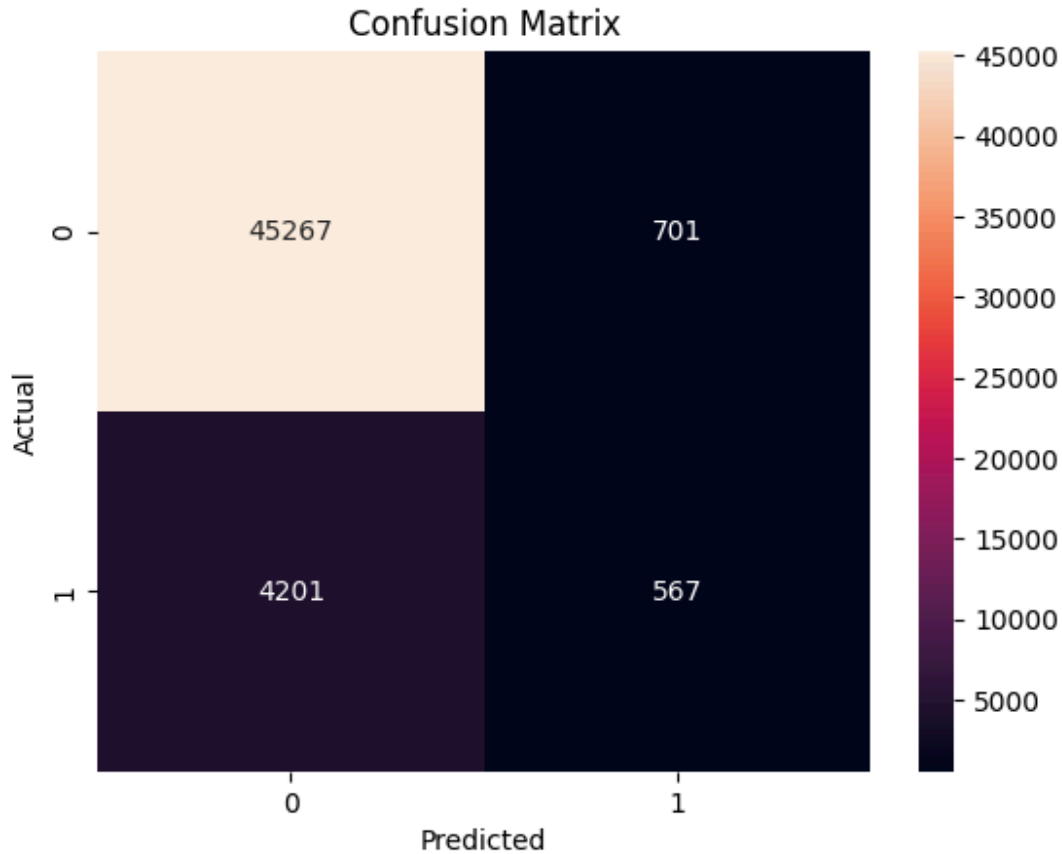
The Random Forest model showed better precision and recall values, indicating its ability to correctly classify both positive and negative cases. These results demonstrate that Random Forest is more effective for heart disease prediction in this dataset.

Figure 2: Feature Importance Analysis of the Machine Learning Model for Heart Disease Prediction



As shown in Figure 2, features such as BMI, Age, and Income have a significant impact on heart disease prediction.

Figure 3: Confusion Matrix of the Random Forest Model for Heart Disease Prediction



As shown in Figure 3, the model demonstrates high accuracy with correct classification of both positive and negative cases.

5. Federated Learning Approach

Federated Learning is a decentralized machine learning technique where multiple clients train models locally on their data without sharing it. Instead of transferring raw data, only model parameters are shared with a central server.

In this work, Federated Learning is proposed as a future enhancement to improve data privacy. By applying this approach, sensitive healthcare data can remain secure while still benefiting from collaborative model training.

6. Conclusion

This paper presented a machine learning-based approach for heart disease prediction using Logistic Regression and Random Forest models. Among the models, Random

Forest achieved the highest accuracy of approximately 95%, making it the most suitable model for this task.

The study also highlights the importance of data privacy and proposes Federated Learning as a future direction. This approach can help build efficient and secure healthcare systems while maintaining high prediction performance.

7. Future Work

Future work can focus on improving the proposed system by integrating advanced deep learning techniques such as neural networks for enhanced prediction accuracy. Additionally, optimizing communication efficiency in federated learning can reduce training time and resource consumption. The use of real-world multi-institutional healthcare datasets can further validate the robustness of the model. Moreover, incorporating advanced privacy techniques such as differential privacy and secure aggregation can strengthen data security and confidentiality.

8. References

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